

Red Sand — play-through review

Theme `redsand` · AP Chemistry, the back half — thermodynamics, equilibrium, kinetics, electrochemistry (grade 12) · 15 sols, ~46 stops · reviewed 2026-08-21 by reading the full book (`books/redsand.yml`), solving every question, and reading opening, stakes, warm-ups and ending in play order.

Verdict

The best *stakes* in the catalogue, achieved with one sentence: everyone here goes home on a rocket that can only leave when the planets line up, it stands four hundred metres away, and it has to lift off full. Nothing else here has a deadline that cannot be argued with, and the ascent vehicle's gauge filling as the campaign advances is the single best piece of world-and-progress binding in the repo.

The chemistry is real and mostly excellent. Water splitting at 237 kJ/mol from $\Delta H - T\Delta S$, Faraday's law from amps to grams of hydrogen, boil-off from a heat leak and a latent heat, freezing-point depression on a perchlorate brine, Beer–Lambert through storm dust, and the recycle-loop distinction between per-pass and overall conversion. The campaign's real subject — announced on the opening card and settled on sol 10 — is that **the thing everyone was arguing about was not the thing that was wrong**: Sundqvist wants the reactor hotter, Herrera says hotter gives back less, and the answer is that a polishing column filled around sol 261 and had been poisoning the catalyst with chloride for thirty-eight sols while everyone read the temperature profile as heat damage. That is the best-constructed misdirection in the set.

One serious problem, and it sits on the campaign's central dispute: **the equilibrium constant for the Sabatier reaction takes three mutually incompatible values across three sols, and one of them is contradicted by arithmetic the player performs themselves**. And the storm ALLOCATE — the first instrument any shipped game authored — can be passed by making no decision at all.

Answerable: 44/46. The two reaction-quotient stops cannot both be right (RS-01), and the ALLOCATE grades a plan nobody had to think about (RS-02). **Sense:** Strong, with the K problem the exception. Every other thread resolves. **Level:** Right for AP Chemistry's back half, and it computes nine of nine syllabus equations. **Fun:** High. A finite tank, a window that does not move, and six people going home on it.

The questions, solved

Verified: $340 \text{ L} \div 85 \text{ L/min} = 4.0 \text{ min residence}$; $286 - 298 \times 0.163 = 237.4 \text{ kJ/mol}$ to split water (the real value is 237.1); $13.8 \times 44.0 \div 16.0 = 37.9 \text{ kg of CO}_2 \text{ a sol}$; $\Delta G = -165,000 - 593 \times (-172) = -63.0 \text{ kJ/mol}$ at the bed's temperature (and the reaction stops wanting to run at $165,000 \div 172 = 959 \text{ K}$, which is the honest answer to that stop's title); $\Delta T_f = 3 \times 1.86 \times 8.0 = 44.6 \text{ K}$ on the brine; $30 \text{ W} \times 88,800 \text{ s} \div 213,000 \text{ J/kg} = 12.5 \text{ kg of oxygen a sol out of the tank}$ (and 88,800 s is a sol to three figures, which I appreciated); $640 \text{ A} \times 88,800 \text{ s} \div 192,970 \text{ C/mol} \times 2.016 \text{ g/mol} = 594 \text{ g of hydrogen a cell a sol}$; $6.9 \text{ kg} \times 55 \text{ kWh/kg} = 380 \text{ kWh}$; $490 \times e^{(-2.4)} = 44 \text{ W/m}^2$ of direct beam through storm dust; $2 \times 32.0 \div 16.0 = 4.0 \text{ kg of oxygen per kg of methane}$.

The sol-9 CONTROL is the best-posed stop here: change the residence time at fixed temperature and pressure, and see whether conversion follows it back on reversal — which is exactly the experiment that

separates a kinetic limit from an equilibrium one, and it is the experiment Herrera's argument needs and never got.

Implemented since this review

- **RS-01, the equilibrium constant.** One enthalpy for the whole campaign: $\Delta H = -123.6$ kJ/mol against $\Delta S = -172$ gives $K = 1,350$ at 533 K, 192 at 573 and 80 at 593 — which is what sols 3 and 6 already claimed to their own rounding, and what the sol-5 ΔG now computes (-21.6 kJ/mol, crossing zero at 720 K). $Q = 282$ is still above the hot ceiling and below the cool one, so Herrera's argument is unchanged and now self-consistent.
- **RS-02, the storm ALLOCATE.** Pool 430 $\rightarrow$ 600 kWh and the drier made a second required answer. Habitat 180 (protected) + liquefier 210 + drier 40 = 430 forced, leaving 170 — which funds the stacks at half rate (160), or the battery reserve with the downlink (140), or the drill with the downlink (110). Three plans, one substantial job each, and the grade now turns on a decision.
- **RS-04, the Q target:** 283 $\rightarrow$ 282.24, the exact product.

Everything else in the findings table below is unchanged, and `npm run check` is green on this theme.

Findings

ID	Sev	Where	Issue	Proposed fix
RS-01	FIX	The Sabatier equilibrium constant — sols 3, 5 and 6	Three incompatible values for one number, and the campaign's own arithmetic refutes two of them. Sol 3 states " <i>K is 190 at 573 K and about 1,050 at 533 K.</i> " Sol 6 states " <i>Q $\approx$ 0.016, which is well below K = 0.14</i> " at 593 K. Sol 5 has the player compute $\Delta G = -63.0$ kJ/mol at 593 K — and $K = \exp(-\Delta G/RT) = \exp(63,000 \div (8.314 \times 593)) = 3.5 \times 10^5$, not 0.14. A 20 K rise cannot move K by a factor of 1,357, and it certainly cannot move it in a way that disagrees with the ΔG the player calculated two sols earlier. Worse, sol 3's conclusion depends entirely on the fabricated value: $Q = 283$ against $K = 190$ gives "the mixture is past the ceiling and the reactor would run backwards", which is the thermodynamic half of the campaign's central argument. Against the true K it is nowhere near the ceiling, and the honest reason a hotter pass gives less is the one the campaign	Two options, and I would take the second. (a) Cheap: make the three agree. Pick one stylised K per temperature, state it in one place, and use it in both stops. This leaves the physics wrong by four orders of magnitude but internally consistent, and no player can catch it without a table of thermodynamic data. (b) Right, and better teaching: keep sol 5's ΔG and let it set K. Then sol 6's comparison becomes $Q = 0.016$ against $K = 3.5 \times 10^5$ — the same forward answer, more emphatically, and the stop's real lesson (Q vs K gives direction, not speed; a poor catalyst can take for ever to close the gap) is <i>strengthened</i> by a huge gap. And sol 3 stops being about an equilibrium ceiling and becomes what it should be: the outlet is far from equilibrium, so the 64% is a kinetic

already teaches on sol 7 — shorter residence time and reverse water-gas-shift selectivity toward CO.

number, and Herrera is right for the reason the sol-9 CONTROL is about to prove rather than the reason he thinks. That change makes the campaign's three-sol argument into one continuous thread instead of two contradictory ones, and it costs one scene, one `why` and one target.

RS- **FIX**
02

Sol 12
ALLOCATE,
"What the
station stops
doing"

The stop can be passed by making no decision. The pool is 430 kWh; habitat heat and air is 180 and `protected: true` ; the liquefier is 210 and its answer is `required: true . instruments.js` sums `protected` picks against the pool, so $180 + 210 = 390$ of **430 committed before the player touches anything**, leaving 40 kWh of discretion — and the grade is `spend ≤ pool && protected in && required answers met` , so **picking nothing else and committing scores full marks**. Every substantial job is unaffordable: stacks 160, batteries 120, drill 90. The only reachable optional item is the drier at exactly 40, or comms at 20 with 20 kWh stranded. The `answerText` promises a decision that does not exist — *"then choose at most one substantial optional job"* — and four of the five answers on the panel go dark whatever the player does.

Restore the trade-off, and make it graded. Raise the storm sol to **560 kWh**: $560 - 390 = 170$, which funds *either* the electrolysis stacks at half rate (160, so the plant makes methane today) *or* the drill, drier and downlink together ($90 + 40 + 20 = 150$, so ice is cut and the cold end stays clear). That is a real choice with a real price and it is exactly the argument the scene sets up. Then add a second `required: true` answer — "Does the cold end stay clear of ice?" is the right one, since letting the drier go cold is how sol 3's blockage happened — so the grade turns on a decision rather than on arithmetic the panel already did. Keep the 430 figure in the *stake* if the drama needs it and let the scene say the liquefier was rescheduled to the night side.

RS- WITHDRAWN
03

Both
SCIENCETANK
stops

This finding was wrong and is withdrawn. I reported that neither tank stop carried `evidence` , having grepped for an `evidence:` key. There is no such book key: `THEME_CONTRACT` 's design is that `rules` is the authorable scoring rule and **the evidence moves up into** `guide` , and the panel drops its own "Evidence available" disclosure when a stop carries `rules` . Both of Red Sand's tanks do exactly that — sol 11's guide gives five facts (38 sols on unpolished water, conductivity logged on every one of them and read by nobody, the array never the limit this season, four probes already in the

Nothing. Recorded because the same wrong finding appears in the Slack Water, Overwind and Dark Fibre reviews and is withdrawn there too. I checked all twenty tank stops in the corpus: every one carries `rules` , and every one I read carries its evidence in the guide.

bed, one charge left with no spare) and its **rules** states the scoring in three numbers with a paragraph explaining each.

RS- 04	WORTH	Sol 3 "Why the hotter pass gave less", target: 283	The arithmetic in the solution gives 282.2 ($0.42 \times 0.7056 \div 0.00105$), and the target is 283. Inside tolerance, so nothing fails — but a player who computes carefully gets a number that is not the one printed.	Set the target to 282 and quote the solution to three figures. Trivial, and worth doing because this is the stop whose exact comparison against K carries the argument.
RS- 05	WORTH	~15 stakes, scenes and verdicts	Numerical normalisation: "where are all 100 carbon moles after 1 pass", " 1 line off the spent charge", "the hot 1 ", " 1 cell", " 1 run", " 1 still". The genuine counts (430 kWh, 6.6 tonnes, sol 261, 38 sols) are intact, and "1 mole" is correct English. The damaged ones are where <i>one</i> was a pronoun or an article.	Editorial pass. Cross-campaign §1. Red Sand is among the less affected, which is consistent with its house style of spelling numbers out.
RS- 06	TASTE	Sol 13 "Three things at once"	The fifth campaign with this mission title and the seventh with a "which of the three cannot wait" stop. Red Sand's version is the best-argued of them — the radiator wins because every sol of doubled boil-off is propellant that leaves for good, which is a genuinely different criterion from the others' — but the title is shared with Blackout, Aftershock, Midway, Quantum and The Trial.	Retitle. Cross-campaign §6.
RS- 07	TASTE	~26 stops (118 collapsed paragraphs)	Repeated format boilerplate in background . Red Sand is the worst-affected book measured: 118 repeated paragraphs, more than any other. Its stop-specific backgrounds are good — "why refrigeration is not negotiable first", "what the answers list is for" — and they are third in a list.	Keep each essay on first use per format. Cross-campaign §2.

Sol-by-sol notes (short)

- **Sols 1–2** — 2,700 kilograms short. Rate against yield stated on the first sol, which is the whole argument, and the drier-and-CO₂-trap CHOICE that explains why the cold end exists before anything freezes in it.
- **Sol 3** — The line that stopped passing gas. The DIAGNOSIS (water past the drier freezing in the exchanger) is well built, and RS-01 starts here.

- **Sols 4–5** — Feed rate, the charge ledger's missing 8%, and the ceiling on running hotter. The catalyst CHOICE — *equilibrium arrives sooner and the mixture settles at the same place* — is the cleanest statement of what a catalyst cannot do anywhere here.
- **Sol 6** — Water is the whole plant. Brine, overpotential as heat, and the Q-versus-K direction call (RS-01).
- **Sols 7–8** — Boil-off, RWGS selectivity, and reading a bed by where its heat is. The exchange-front CHOICE (the outlet rises toward the inlet concentration as the front reaches the end) is the mechanism sol 10 needs.
- **Sols 9–10** — The assay bench and the trace nobody read. **The best two sols in the campaign.** The chloride trace's shape says the column filled around sol 261 and has passed feed straight through since, and the CONTROL then separates kinetics from equilibrium properly. This is the payoff and it lands.
- **Sols 11–12** — A new charge, a rule for the water, and the storm. RS-02 and RS-03 sit here.
- **Sols 13–15** — The radiator, loading, and what the next crew is told. The TRIGGER — write the rule before the trace moves — is the right last instrument for a campaign whose whole failure was a trace nobody read.

Opening and closing

Opening: "Everyone here goes home on a rocket that can only leave when Mars and Earth line up. The next alignment is sixteen months off. It stands on the pad four hundred metres away and has to lift off full. Nothing is flown from Earth to fill it... So far the plant has made 3.9." Deadline, mechanism, authority, and both sides of the argument. This is the rewritten card and it is now one of the two or three best in the set.

Closing: three paragraphs and all three do work. The middle one names the cost honestly — eleven sols of production, one catalyst charge with no replacement on the planet, four hundred kilograms of oxygen out of the vent — and the unfinished list is the best in the repo: a change interval that is a guess taken from one breakthrough, an alarm wired four sols before the rotation ended that has never sounded, and nobody knowing what the deeper ice does to a catalyst. The last paragraph is addressed to the player. Keep all of it.

Warm-ups

All seven authored and specific to a buried plant on a track. No findings.

What to keep

- The window. A deadline set by orbital mechanics is the only deadline in this catalogue that cannot be negotiated, and the campaign never lets anyone try.
- The ascent vehicle's gauge. Progress the player can see from anywhere on the site.
- Sundqvist and Herrera arguing about a set point for a fortnight while the actual fault is a filled column upstream of both of them.
- "You found a poisoned catalyst that everybody else was reading as heat damage."